

Arbitrage-free SVI volatility surfaces

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Abstract

In this article, we show how to calibrate the widely-used SVI parameterization of the implied volatility smile in such a way as to guarantee the absence of static arbitrage. In particular, we exhibit a large class of arbitrage-free SVI volatility surfaces with a simple closed-form representation. We demonstrate the high quality of typical SVI fits with a numerical example using recent SPX options data.

1 Introduction

The *stochastic volatility inspired* or *SVI* parameterization of the implied volatility smile was originally devised at Merrill Lynch in 1999 and subsequently publicly disseminated in [13]. This parameterization has two key properties that have led to its popularity with practitioners:

- For a fixed time to expiry t , the implied Black-Scholes variance $\sigma_{BS}^2(k, t)$ is linear in the log-strike k as $|k| \rightarrow \infty$ consistent with Roger Lee's moment formula [23].
- It is relatively easy to fit listed option prices whilst ensuring no calendar spread arbitrage.

The consistency of the SVI parameterization with arbitrage bounds for extreme strikes has also led to its use as an extrapolation formula [20]. Moreover, as shown in [15], the SVI parameterization is not arbitrary in the sense that the large-maturity limit of the Heston implied volatility smile is exactly SVI. However it is well-known that SVI smiles may be arbitrageable. Previous work has shown how to calibrate SVI to given implied volatility data (for example [27]). Other recent work [6] has been concerned with showing

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Part 2 — Now the full Black-Scholes reduction, every algebraic step shown explicitly

Original formula (standard Black-Scholes-Merton, textbook notation S_0, K, T, r, q, σ):

$$C = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2), \quad d_1 = \frac{\ln(S_0/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

Substitution: $S_0 = F_T e^{-(r-q)T}$ (from Part 1). Substitute into $\ln(S_0/K)$ by hand:

$$\ln(S_0/K) = \ln\left(\frac{F_T e^{-(r-q)T}}{K}\right) = \ln(F_T/K) + \ln\left(e^{-(r-q)T}\right) = \ln(F_T/K) - (r - q)T.$$

Substitute into d_1 :

$$d_1 = \frac{[\ln(F_T/K) - (r - q)T] + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}} = \frac{\ln(F_T/K) - (r - q)T + (r - q)T + \frac{\sigma^2}{2}T}{\sigma\sqrt{T}}.$$

The $-(r - q)T$ and $+(r - q)T$ terms cancel exactly, leaving:

$$d_1 = \frac{\ln(F_T/K) + \sigma^2 T/2}{\sigma\sqrt{T}}.$$

(Confirmed symbolically above: $\sigma^2 T/2 + \log(F/K)$, no r, q remaining.) Then

$$d_2 = d_1 - \sigma\sqrt{T} = \frac{\ln(F_T/K) + \sigma^2 T/2}{\sigma\sqrt{T}} - \sigma\sqrt{T} = \frac{\ln(F_T/K) + \sigma^2 T/2 - \sigma^2 T}{\sigma\sqrt{T}} = \frac{\ln(F_T/K) - \sigma^2 T/2}{\sigma\sqrt{T}}.$$

Now the prefactor of $N(d_1)$, which was $S_0 e^{-qT}$:

$$S_0 e^{-qT} = F_T e^{-(r-q)T} \cdot e^{-qT} = F_T e^{-(r-q)T - qT} = F_T e^{-rT + qT - qT} = F_T e^{-rT}.$$

This now shares the identical factor e^{-rT} with the second term's own $K e^{-rT}$ (confirmed symbolically above: $F \cdot \exp(-T \cdot r)$). So:

$$C = F_T e^{-rT} N(d_1) - K e^{-rT} N(d_2) = e^{-rT} [F_T N(d_1) - K N(d_2)].$$

should it occur. With both types of arbitrage that typically calibrates well to given implied static arbitrage

Under the risk-neutral measure, S_t solves the SDE

$$dS_t = (r - q) S_t dt + \sigma S_t dW_t,$$

with W_t a standard Brownian motion under that measure. Solving via Itô's lemma on $\ln S_t$ gives the standard GBM solution:

$$S_t = S_0 \exp\left[\left(r - q - \frac{\sigma^2}{2}\right)t + \sigma W_t\right].$$

Now take the expectation, conditional on $\mathcal{F}_0$ (i.e., S_0 known):

$$F_t := \mathbb{E}[S_t \mid \mathcal{F}_0] = S_0 e^{(r-q-\sigma^2/2)t} \cdot \mathbb{E}[e^{\sigma W_t}].$$

Since $W_t \sim N(0, t)$ under this measure, $\sigma W_t \sim N(0, \sigma^2 t)$, and the moment generating function of a centered Gaussian gives $\mathbb{E}[e^{\sigma W_t}] = e^{\sigma^2 t/2}$ (confirmed numerically above). Substituting:

$$F_t = S_0 e^{(r-q-\sigma^2/2)t} \cdot e^{\sigma^2 t/2} = S_0 e^{(r-q)t}.$$

ing, we consider a stock price process $(S_t)_{t \geq 0}$ with natural filtration $(\mathcal{F}_t)_{t \geq 0}$, and we define the forward price process $(F_t)_{t \geq 0}$ by $F_t := \mathbb{E}(S_t | \mathcal{F}_0)$. For any $k \in \mathbb{R}$ and $t > 0$, $C_{BS}(k, \sigma^2 t)$ denotes the Black-Scholes price of a European Call option on S with strike $K = F_t e^k$, maturity t and volatility $\sigma > 0$. We shall denote the Black-Scholes implied volatility by $\sigma_{BS}(k, t)$, and define the total implied variance by

$$w(k, t) = \sigma_{BS}^2(k, t)t.$$

The implied variance v shall be equivalently defined as $v(k, t) = \sigma_{BS}^2(k, t) = w(k, t)/t$. We shall refer to the two-dimensional map $(k, t) \mapsto w(k, t)$ as the volatility surface, and for any fixed maturity $t > 0$, the function $k \mapsto w(k, t)$ will represent a slice. We propose below three different—yet equivalent—slice parameterizations of the total implied variance, and give the exact correspondence between them. For a given maturity slice, we shall use the notation $w(k; \chi)$ where χ represents a set of parameters, and drop the t -dependence.

Introduce $w := \sigma^2 T$ (the paper's total variance): $\sigma\sqrt{T} = \sqrt{\sigma^2 T} = \sqrt{w}$, so

$$d_1 = \frac{\ln(F_T/K) + w/2}{\sqrt{w}}, \quad d_2 = \frac{\ln(F_T/K) - w/2}{\sqrt{w}}.$$

Introduce $k := \ln(K/F_T)$ (the paper's log-moneyness), i.e. $K = F_T e^k \Rightarrow \ln(F_T/K) = -k$:

$$d_1 = \frac{-k + w/2}{\sqrt{w}} = -\frac{k}{\sqrt{w}} + \frac{\sqrt{w}}{2} =: d_+(k), \quad d_2 = \frac{-k - w/2}{\sqrt{w}} = -\frac{k}{\sqrt{w}} - \frac{\sqrt{w}}{2} =: d_-(k),$$

matching the paper's stated $d_{\pm}(k)$ exactly, letter for letter.

Substitute $K = F_T e^k$ into the bracket:

$$F_T N(d_1) - K N(d_2) = F_T N(d_+(k)) - F_T e^k N(d_-(k)) = F_T \{N(d_+(k)) - e^k N(d_-(k))\}.$$

Final result

$$C = e^{-rT} C_{BS}(k, w), \quad C_{BS}(k, w) := F_T \{N(d_+(k)) - e^k N(d_-(k))\}, \quad w = \sigma^2 T, \quad k = \ln(K/F_T).$$

that the corresponding total variance must

Set $r = 0$: $e^{-rT} = e^0 = 1$, so

$$C = C_{BS}(k, w) \quad \text{exactly.}$$

No condition needed on q

This holds regardless of the dividend yield q — dividends never leave the picture, because q only enters through the forward $F_T = S_0 e^{(r-q)T}$, and F_T is still fully present inside $C_{BS}(k, w)$ (via $k = \ln(K/F_T)$). Setting $r = 0$ only kills the discount factor e^{-rT} , which is a separate multiplicative piece from the dividend-adjustment baked into F_T . So even with $r = 0$ and $q \neq 0$, $C = C_{BS}$ still holds — you'd just have $F_T = S_0 e^{-qT}$ in that case.

The formula

$$\frac{C_{BS}(k, w)}{F_T} = N(d_+(k)) - e^k N(d_-(k))$$

is a universal function of (k, w) alone — it has zero dependence on q, r , or S_0 individually. The second block above proves this numerically: at the exact same $k = 0.10$, the normalized price C_{BS}/F comes out to **0.079770** regardless of q — identical to the full displayed precision. So the "same formula regardless" intuition is correct, but only for the normalized price (price per unit forward) as a function of (k, w) . It is not true for the real dollar price at a fixed real strike K — because converting between "fixed real K " and "fixed k " requires F_T , and F_T is exactly where q (and r) reenter.

the absence of static arbitrage q never disappears; it gets fully absorbed into two places, both involving F_T :

- Computing k from a real strike: $k = \ln(K/F_T)$ — since $F_T = S_0 e^{(r-q)T}$ depends on q , the same dollar strike $K = 100$ corresponds to a different k depending on q (confirmed above: k goes from 0 to 0.03 to 0.08 as q rises).

2 Characterisation of static arbitrage

In this section we provide model-independent definitions of (static) arbitrage and some preliminary results. We define static arbitrage for a given volatility surface in the following way, which is equivalent to the definition of static arbitrage for call options recalled in the introduction (see also [25]).

Definition 2.1. *A volatility surface is free of static arbitrage if and only if the following conditions are satisfied:*

- (i) *it is free of calendar spread arbitrage;*
- (ii) *each time slice is free of butterfly arbitrage.*

In particular, absence of butterfly arbitrage ensures the existence of a (non-negative) probability density, and absence of calendar spread arbitrage implies monotonicity of option prices with respect to the maturity. The following two subsections analyse in details each of these two types of arbitrage, in a model-independent way.

2.1 Calendar spread arbitrage

Calendar spread arbitrage is usually expressed in terms of option prices with respect to the maturity (see for example [11] and [13]). If your model ever priced the 2-month option cheaper than the 1-month one (at matched money), you'd have found free money: sell the overpriced 1-month option, buy the underpriced 2-month option, and — because the 2-month option's payoff dominates the 1-month one's in every scenario — you can never lose, and you might gain if the market reverts. That's the arbitrage: a trade with no downside and positive upside, funded by a pricing error.

Calendar spread arbitrage

Financially: Imagine two call options on the same stock, same moneyness (say, both are 5% out-of-the-money), but one expires in 1 month and the other in 2 months. Common sense says the 2-month option should be worth at least as much as the 1-month one — it gives you everything the 1-month option gives you, plus an extra month of optionality. It literally cannot be worth less.

Lemma 2.1. *If dividends are proportional to the stock price, the volatility surface w is free of calendar spread arbitrage if and only if*

$$\partial_t w(k, t) \geq 0, \quad \text{for all } k \in \mathbb{R} \text{ and } t > 0.$$

In plain terms: if you plot the total-variance curve $k \mapsto w(k, t)$ for two different maturities on the same graph, the longer-dated curve must sit entirely above (or touching) the shorter-dated curve — the curves must never cross. If they cross, that crossing point is exactly where the arbitrage trade above becomes available.

Proof. Let $(X_t)_{t \geq 0}$ be a martingale, $L \geq 0$ and $0 \leq t_1 < t_2$. Then the inequality

$$\mathbb{E}[(X_{t_2} - L)^+] \geq \mathbb{E}[(X_{t_1} - L)^+]$$

is standard. For any $i = 1, 2$, let C_i be options with strikes K_i and expirations t_i . Suppose that the two options have the same moneyness, i.e.

$$\frac{K_1}{F_{t_1}} = \frac{K_2}{F_{t_2}} =: e^k$$

Then, if dividends are proportional, the process $(X_t)_{t \geq 0}$ defined by $X_t := S_t/F_t$ for all $t \geq 0$ is a martingale and

$$\frac{C_2}{K_2} = e^{-k} \mathbb{E}[(X_{t_2} - e^k)^+] \geq e^{-k} \mathbb{E}[(X_{t_1} - e^k)^+] = \frac{C_1}{K_1}$$

So, if dividends are proportional, keeping the moneyness constant, option prices are non-decreasing in time to expiration. The Black-Scholes formula for the non-discounted value of an option may be expressed in the form $C_{\text{BS}}(k, w(k, t))$ with C_{BS} strictly increasing in its second argument. It follows that for fixed k , the function $w(k, \cdot)$ must be non-decreasing. $\square$

Lemma 2.1 motivates the following definition.

Definition 2.2. A volatility surface w is free of calendar spread arbitrage if

$$\partial_t w(k, t) \geq 0, \quad \text{for all } k \in \mathbb{R} \text{ and } t > 0.$$

Butterfly arbitrage

Financially: This one is about a single expiry, not two. A "butterfly spread" is a trade built from three strikes: buy one call at a low strike, sell two calls at a middle strike, buy one call at a high strike (equally spaced). This payoff is always ≥ 0 at expiry, and strictly positive if the stock lands near the middle strike. So the butterfly's price today must also be ≥ 0 — you can't pay negative money for a portfolio that can only pay off zero or more.

The price of a very tightly spaced butterfly (strikes close together) is essentially measuring the local probability density of the stock landing near that strike. If your model's implied volatility smile is shaped in a way that makes this tiny butterfly cost negative, that means the model is implicitly assigning a **negative probability** to some outcome — which is nonsensical, and also exploitable: you can buy that negatively-priced butterfly (getting paid to hold it) and its payoff is never negative, so it's free money with no downside.

to be free of calendar spread arbitrage. We now consider a different type of arbitrage, namely butterfly arbitrage (Definition 2.3). Absence of this arbitrage corresponds to the existence of a risk-neutral martingale measure and the classical definition of no static arbitrage, as developed in [12] or [8]. In this section, we consider only one slice of the implied volatility surface, i.e. the map $k \mapsto w(k, t)$ for a given fixed maturity $t > 0$. For clarity we therefore drop—in this section only—the t -dependence of the smile and use the notation $w(k)$ instead. Unless otherwise stated, we shall always assume that the map $k \mapsto w(k, t)$ is at least of class $\mathcal{C}^2(\mathbb{R})$ for all $t \geq 0$.

Definition 2.3. A slice is said to be free of butterfly arbitrage if the corresponding density is non-negative.

Recall the Black-Scholes formula for a European call option price:

$$C_{\text{BS}}(k, w(k)) = F_t(\mathcal{N}(d_+(k)) - e^k \mathcal{N}(d_-(k))), \quad \text{for all } k \in \mathbb{R},$$

where $\mathcal{N}$ is the Gaussian cdf and $d_{\pm}(k) := -k/\sqrt{w(k)} \pm \sqrt{w(k)}/2$. Let us define the function $g : \mathbb{R} \rightarrow \mathbb{R}$ by

$$g(k) := \left(1 - \frac{k w'(k)}{2 w(k)}\right)^2 - \frac{w'(k)^2}{4} \left(\frac{1}{w(k)} + \frac{1}{4}\right) + \frac{w''(k)}{2}. \quad (2.1)$$

This function will be the main ingredient in the determination of butterfly arbitrage as stated in the following lemma.

Lemma 2.2. A slice is free of butterfly arbitrage if and only if $g(k) \geq 0$ for all $k \in \mathbb{R}$ and $\lim_{k \rightarrow +\infty} d_+(k) = -\infty$.

$$p(k) = \frac{\partial^2 C(k, t)}{\partial K^2}.$$

A "well-behaved" (arbitrage-free) smile needs $p(k) \geq 0$ everywhere — a proper density can't go negative. Working this through the Black-Scholes formula in terms of $w(k)$ and its derivatives $w'(k)$, $w''(k)$, this condition reduces to a single function

needing to satisfy $g(k) \geq 0$ for every k (plus a tail condition so the implied density actually integrates to 1). If $g(k)$ dips below zero anywhere, the model is implying a negative density there — and a butterfly spread struck around that point is mispriced, giving you the arbitrage.

Proof. It is well known [2] that the probability density function p may be computed from the call price function C as

$$p(k) = \frac{\partial^2 C(k)}{\partial K^2} \Big|_{K=F_t e^k} = \frac{\partial^2 C_{BS}(k, w(k))}{\partial K^2} \Big|_{K=F_t e^k}, \quad \text{for any } k \in \mathbb{R}.$$

Explicit differentiation of the Black-Scholes formula then gives for any $k \in \mathbb{R}$,

$$p(k) = \frac{g(k)}{\sqrt{2\pi w(k)}} \exp\left(-\frac{d_-(k)^2}{2}\right).$$

We have so far implicitly assumed that the function p is a well-defined density, and in particular that it integrates to one. This may not always be the case though, and one needs to impose asymptotic boundary conditions. In particular, call prices must converge to 0 as k tends to infinity, which is equivalent to having $\lim_{k \rightarrow +\infty} d_+(k) = -\infty$. We refer the reader to [24] for a proof of this equivalence. $\square$

3 SVI formulations

Section 3 of the paper presents three equivalent ways of parameterizing a single implied volatility smile (fixed maturity t), each suited to a different purpose — tractability, geometric intuition, or trader interpretability. Let me go through each.

We first recall here the original SVI formulation proposed in [13], and then present some alternative (but equivalent) ones. We emphasize in particular that even though the original (“raw”) formulation is very tractable and has become popular with practitioners, it is difficult—seemingly impossible—to find precise conditions on the parameters to prevent arbitrage.

****Why it works asymptotically:**** as $k \rightarrow \pm\infty$, $\sqrt{(k-m)^2 + \sigma^2} \sim |k-m|$, so

$$w(k) \sim a + b(1 \pm \rho)|k| \quad \text{as } k \rightarrow \pm\infty,$$

3.1 The raw SVI parameterization

which is linear in $|k|$ — exactly the behavior required by Roger Lee's moment formula for the wings of a legitimate volatility smile. This is *the* structural reason SVI was designed

For a given parameter set $\chi_R = \{a, b, \rho, m, \sigma\}$, the raw SVI parameterization of total implied variance reads:

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$$w(k; \chi_R) = a + b \left\{ \rho(k-m) + \sqrt{(k-m)^2 + \sigma^2} \right\}, \quad (3.1)$$

where $a \in \mathbb{R}$, $b \geq 0$, $|\rho| < 1$, $m \in \mathbb{R}$, $\sigma > 0$, and the obvious condition $a + b\sigma\sqrt{1-\rho^2} \geq 0$, which ensures that $w(k; \chi_R) \geq 0$ for all $k \in \mathbb{R}$. This condition indeed ensures that the minimum of the function $w(\cdot; \chi_R)$ is non-negative. Note further that the function $k \mapsto w(k; \chi_R)$ is (strictly) convex on the whole real line. It follows immediately that changes in the parameters have the following effects:

- Increasing a increases the general level of variance, a vertical translation of the smile;
- Increasing b increases the slopes of both the put and call wings, tightening the smile;
- Increasing ρ decreases (increases) the slope of the left(right) wing, a counter-clockwise rotation of the smile;

Why it's popular but flawed: raw SVI is very tractable for fitting (5 parameters, closed form, well-behaved asymptotics), but as the earlier discussion showed, there's no known simple closed-form condition on (a, b, ρ, m, σ) that guarantees $g(k) \geq 0$ everywhere — hence real raw-SVI fits (like the Vogt example) can and do violate no-arbitrage.

- Increasing m translates the smile to the right;
- Increasing σ reduces the at-the-money (ATM) curvature of the smile.

We exclude the trivial cases $\rho = 1$ and $\rho = -1$, where the volatility smile is respectively strictly increasing and decreasing. We also exclude the case $\sigma = 0$ which corresponds to a linear smile.

and the inverse transform is given symmetrically. So "natural" SVI carries exactly the same information, just re-parameterized — the point of introducing it is that it's the form that generalizes cleanly to **SSVI** later (Section 4), because it separates out $\Delta, \mu, \omega, \zeta$ in a way that lets you promote ω (and implicitly, level/curvature) to be functions of maturity t via the ATM total variance θ_t .

3.2 The natural SVI parameter

For a given parameter set $\chi_N = \{\Delta, \mu, \rho, \omega, \zeta\}$, the *natural SVI parameterization* of **total implied variance** reads:

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$$w(k; \chi_N) = \Delta + \frac{\omega}{2} \left\{ 1 + \zeta \rho (k - \mu) + \sqrt{(\zeta(k - \mu) + \rho)^2 + (1 - \rho^2)} \right\}, \quad (3.2)$$

where $\omega \geq 0$, $\Delta \in \mathbb{R}$, $\mu \in \mathbb{R}$, $|\rho| < 1$ and $\zeta > 0$. It is straightforward to derive the following correspondence between the raw and natural SVI parameters:

Lemma 3.1. *We have the following mapping of parameters between the raw and the natural SVI:*

$$(a, b, \rho, m, \sigma) = \left(\Delta + \frac{\omega}{2} (1 - \rho^2), \frac{\omega \zeta}{2}, \rho, \mu - \frac{\rho}{\zeta}, \frac{\sqrt{1 - \rho^2}}{\zeta} \right), \quad (3.3)$$

and its inverse transformation, between the natural and the raw SVI:

$$(\Delta, \mu, \rho, \omega, \zeta) = \left(a - \frac{\omega}{2} (1 - \rho^2), m + \frac{\rho \sigma}{\sqrt{1 - \rho^2}}, \rho, \frac{2b\sigma}{\sqrt{1 - \rho^2}}, \frac{\sqrt{1 - \rho^2}}{\sigma} \right). \quad (3.4)$$

3.3 The SVI Jump-Wings (SVI-JW) parameterization

Neither the raw SVI nor the natural SVI parameterizations are intuitive to traders in the sense that a trader cannot be expected to carry around the typical value of these parameters in his head. Moreover, there is no reason to expect these parameters to be particularly stable. The *SVI-Jump-Wings (SVI-JW) parameterization* of the implied variance v (rather than the implied total variance w) was inspired by a similar parameterization attributed to Tim Klassen, then at Goldman Sachs. For a given time to expiry $t > 0$ and a parameter set $\chi_J = \{v_t, \psi_t, p_t, c_t, \tilde{v}_t\}$ the SVI-JW parameters are defined from

Unlike raw/natural SVI, this parameterizes the **implied variance** (not total variance), and crucially the parameters have direct trading-desk meaning:

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Recall from (3.5) that v_t is the **ATM implied variance**:

$$v_t = \frac{a + b\{-\rho m + \sqrt{m^2 + \sigma^2}\}}{t}.$$

This is $\sigma_{BS}^2(0, t)$ — literally the squared implied vol at the money, for maturity t .

Then $w_t := v_t t$ undoes the division by t , giving you back

$$w_t = a + b\{-\rho m + \sqrt{m^2 + \sigma^2}\} = w(0; \chi_R),$$

which is exactly the **ATM total implied variance** — the raw SVI curve $w(k)$ evaluated at $k = 0$.

the **raw SVI parameters** as follows:

$$\begin{aligned} v_t &= \frac{a + b \{-\rho m + \sqrt{m^2 + \sigma^2}\}}{t}, \\ \psi_t &= \frac{1}{\sqrt{w_t}} \frac{b}{2} \left(-\frac{m}{\sqrt{m^2 + \sigma^2}} + \rho \right), \\ p_t &= \frac{1}{\sqrt{w_t}} b (1 - \rho), \\ c_t &= \frac{1}{\sqrt{w_t}} b (1 + \rho), \\ \tilde{v}_t &= \frac{1}{t} \left(a + b \sigma \sqrt{1 - \rho^2} \right), \end{aligned} \tag{3.5}$$

with $w_t := v_t t$. Note that this parameterization has an explicit dependence on the time to expiration t , and hence can be viewed as generalizing the raw (expiration-independent) SVI parameterization. The SVI-JW parameters have the following interpretations:

- v_t — **ATM implied variance (the level of the smile at $k = 0$)**
- ψ_t — **ATM skew: literally $\psi_t = \partial_k \sigma_{BS}(k, t)|_{k=0}$, the slope of the vol curve at the money**
- p_t — **slope of the left (put) wing as $k \rightarrow -\infty$**
- c_t — **slope of the right (call) wing as $k \rightarrow +\infty$**
- $\tilde{v}_t$ — **the minimum variance value the smile ever attains**

If smiles scaled perfectly as $1/\sqrt{w_t}$ (as is approximately the case empirically), these parameters would be constant, independent of the slice t . This makes it easy to extrapolate the SVI surface to expirations beyond the longest expiration in the data set. Also note that by definition, for any $t > 0$ we have

$$\psi_t = \left. \frac{\partial \sigma_{BS}(k, t)}{\partial k} \right|_{k=0}$$

The choice of volatility skew as the skew measure rather than variance skew for example, reflects the empirical observation that volatility is roughly lognormally distributed. Specifically, following the lines of [14, Chapter 7], assume that the **instantaneous variance process satisfies the SDE**

$$dv_t = \alpha(v_t) dt + \eta \sqrt{v_t} \beta(v_t) dZ_t, \quad \text{for all } t \geq 0$$

where $\eta > 0$, $(Z_t)_{t \geq 0}$ is a standard Brownian motion and α and β two functions on $\mathbb{R}_+$ ensuring the existence of a unique strong solution to the SDE (see for instance [22] for exact conditions), then the **ATM variance skew**

$$\lim_{t \rightarrow 0} \left. \frac{\partial \sigma_{BS}(k, t)^2}{\partial k} \right|_{k=0}$$

Why this matters practically: a trader can look at these numbers and immediately understand the smile's shape, whereas raw parameters (a, b, ρ, m, σ) have no such direct trading interpretation. The paper also notes the identity $\psi_t = \partial_k \sigma_{BS}(k, t)|_{k=0}$ has a theoretical justification: if instantaneous variance follows an SDE $dv_t = \alpha(v_t)dt + \eta\sqrt{v_t}\beta(v_t)dZ_t$ with $\beta(v) \sim \sqrt{v}$ (lognormal-like vol-of-vol), then the ATM volatility skew (not variance skew) converges to a constant as $t \rightarrow 0$ — matching empirical behavior of vol dynamics. That's the mathematical reason vol-skew, rather than variance-skew, was chosen as the natural JW quantity.

exists and is proportional to $\beta(v)$. If the variance process is lognormal so that $\beta(v)$ behaves like $\sqrt{v}$, the limit of the at-the-money *volatility* skew as time to expiry tends to zero is constant and independent of the volatility level. This consistency of the SVI-JW parameterization with empirical volatility dynamics thus leads in practice to greater parameter stability over time. The following lemma provides the inverse representation of (3.5).

Lemma 3.2. Assume that $m \neq 0$. For any $t > 0$, define the (t -dependent) quantities:

$$\beta := \rho - \frac{2\psi_t\sqrt{w_t}}{b} \quad \text{and} \quad \alpha := \text{sign}(\beta)\sqrt{\frac{1}{\beta^2} - 1}.$$

where we have further assumed that $\beta \in [-1, 1]^1$. Then, the raw SVI and SVI-JW parameters are related as follows:

$$\begin{aligned} b &= \frac{\sqrt{w_t}}{2} (c_t + p_t), \\ \rho &= 1 - \frac{p_t \sqrt{w_t}}{b}, \\ a &= \tilde{v}_t t - b\sigma\sqrt{1 - \rho^2}, \\ m &= \frac{(v_t - \tilde{v}_t) t}{b \left\{ -\rho + \text{sign}(\alpha)\sqrt{1 + \alpha^2} - \alpha\sqrt{1 - \rho^2} \right\}}, \\ \sigma &= \alpha m. \end{aligned}$$

If $m = 0$, then the formulae above for b , ρ and a still hold, but $\sigma = (v_t t - a)/b$.

Proof. The expressions for b , ρ and a follow directly from (3.5). Assume that $m \neq 0$ and let $\beta := \rho - 2\psi_t\sqrt{w_t}/b$ and $\alpha := \sigma/m \in \mathbb{R}$. Then the expressions in (3.5) give

$$\beta = \frac{\text{sign}(\alpha)}{\sqrt{1 + \alpha^2}},$$

which implies that

$$\alpha = \text{sign}(\beta)\sqrt{\frac{1}{\beta^2} - 1}.$$

Using (3.5), we also have

$$\frac{(v_t - \tilde{v}_t) t}{b} = m \left\{ -\rho + \text{sign}(\alpha)\sqrt{1 + \alpha^2} - \alpha\sqrt{1 - \rho^2} \right\},$$

from which we deduce m in terms of α , and the expression of σ is recovered from the equality $\sigma = \alpha m$. The expression for σ in the case $m = 0$ is straightforward from (3.5). $\square$

¹The condition $\beta \in [-1, 1]$ is equivalent to $-p_t \leq 2\psi_t < c_t$, i.e. to the convexity of the smile.

All three describe the exact same curve for a fixed t — the paper's Lemma 3.1 and Lemma 3.2 give the exact bijective maps between them — so the choice of formulation is purely about which coordinates are most convenient for a given task: fitting (raw), extending to a surface (natural), or interpreting/calibrating against market-observable quantities (SVI-JW). This is exactly why Section 4's SSVI construction is built on top of natural SVI, while Section 5's calibration algorithm exploits SVI-JW parameters (in particular c_t and $\tilde{v}_t$) to repair butterfly arbitrage in a way that's interpretable and tractable.

A volatility **surface** is a function of two variables: $w(k, t)$ — it tells you the total variance for any strike k and any maturity t . Think of it as an actual 2D sheet sitting over the (k, t) -plane.

A **slice** is what you get by freezing t at one value and looking only at $k \mapsto w(k, t)$ — a single curve, like taking a knife and cutting a cross-section through the sheet at one maturity.

Why raw SVI is only a slice model

Look at the raw SVI formula again:

$$w(k; \chi_n) = a + b \left\{ \rho(k - m) + \sqrt{(k - m)^2 + \sigma^2} \right\}.$$

There's no t anywhere in this formula. It's a curve in k only. So raw SVI, by itself, doesn't tell you how the smile evolves as maturity changes — It just gives you "one" curve.

To get a full surface, in practice what people do (and what the paper assumes) is: fit a ***separate* raw SVI curve to the market data at each observed expiry**. So if your data has expiries $t_1, t_2, \dots, t_n$ (say, 1 week, 1 month, 3 months, ...), you end up with n ***independent* parameter sets**:

$$\chi_1 = \{a_1, b_1, \rho_1, m_1, \sigma_1\} \text{ for } t_1, \quad \chi_2 = \{a_2, b_2, \rho_2, m_2, \sigma_2\} \text{ for } t_2, \quad \dots$$

Stack these curves side by side (one per maturity) and that collection ***is*** your "surface" — but it's a ****discrete**** stack of curves, not a single smooth 2-variable formula. There's no built-in rule connecting χ_1 to χ_2 ; they were fit independently.

3.4 Arbitrage and absence thereof in SVI parameterizations

Given a volatility surface, it is natural to wonder whether it is free of arbitrage. Since we can easily switch from any of the SVI formulations to either of the other two using Lemma 3.3 and Lemma 3.2, we shall state the following results only for the raw SVI parameterization (3.1). Referring to (3.1) as a volatility surface is a slight abuse of language since (3.1) is really an expiry-independent slice parameterization. A volatility surface is thus understood as a (discrete) collection of slices, with a different set of parameters for each expiry. Checking calendar arbitrage in the sense of Lemma 2.1 is then equivalent to checking for calendar arbitrage for any pair of expiries t_1 and t_2 . The following lemma establishes a sufficient condition for the absence of calendar spread arbitrage.

Lemma 3.3. *The raw SVI surface (3.1) is free of calendar spread arbitrage if a certain quartic polynomial (given in (3.7) below) has no real root.*

Proof. By definition, there is no calendar arbitrage if for any two dates $t_1 \neq t_2$, the corresponding slices $w(\cdot, t_1)$ and $w(\cdot, t_2)$ do not intersect. Let these two slices be characterised by the sets of parameters $\chi_1 := \{a_1, b_1, \sigma_1, \rho_1, m_1\}$ and $\chi_2 := \{a_2, b_2, \sigma_2, \rho_2, m_2\}$, and assume for convenience that $0 < t_1 < t_2$. We therefore need to determine the (real) roots of the equation $w(k, t_1) = w(k, t_2)$. The latter is equivalent to

$$a_1 + b_1 \left\{ \rho_1 (k - m_1) + \sqrt{(k - m_1)^2 + \sigma_1^2} \right\} = a_2 + b_2 \left\{ \rho_2 (k - m_2) + \sqrt{(k - m_2)^2 + \sigma_2^2} \right\}. \quad (3.6)$$

Leaving $\sqrt{(k - m_1)^2 + \sigma_1^2}$ on one side, squaring the equality and rearranging it leads to

$$2b_2 (\alpha + \beta k) \sqrt{(k - m_2)^2 + \sigma_2^2} = b_1^2 \{ (k - m_1)^2 + \sigma_1^2 \} - b_2^2 \{ (k - m_2)^2 + \sigma_2^2 \} - (\alpha + \beta k)^2,$$

where $\alpha := a_2 - a_1 + b_1 \rho_1 m_1 - b_2 \rho_2 m_2$ and $\beta := b_2 \rho_2 - b_1 \rho_1$. Squaring the last equation above gives a quartic polynomial equation of the form

$$\alpha_4 k^4 + \alpha_3 k^3 + \alpha_2 k^2 + \alpha_1 k + \alpha_0 = 0, \quad (3.7)$$

k is the unknown. It is the log-strike — a real-valued variable — and you're finding the specific real values of k (if any) where the two smiles intersect. It's not "linear," it's degree 4, because two square-roots got squared away in the derivation.

where each of the coefficients lengthy yet explicit expressions² in terms of the parameters $\{a_1, b_1, \rho_1, \sigma_1, m_1\}$ and $\{a_2, b_2, \rho_2, \sigma_2, m_2\}$. If this quartic polynomial has no real root, then the slices do not intersect and the lemma follows. Roots of a quartic polynomial are known in closed-form thanks to Ferrari and Cardano [3]. Thus there exist closed-form expressions in terms of χ_1 and χ_2 for the possible intersection points of the two SVI slices. $\square$

Remark 3.1. *If the quartic polynomial (3.7) has one or more real roots, we need to check whether the latter are indeed solutions of the original problem (3.6), or spurious solutions arising from the two squaring operations. The absence of real roots of the quartic polynomial is clearly a sufficient—but not necessary—condition.*

²Explicit expressions for these coefficients can be found in the baruch.cuny.edu/jgatheral.

Caveat (Remark 3.1). This is a **sufficient, not necessary** condition, for a subtle reason: squaring an equation can introduce *extraneous* roots — solutions of the squared equation that don't solve the original unsquared equation (3.6). So:

- No real roots of the quartic $\implies$ no crossing (valid, clean implication).
- Real roots exist $\implies$ you must separately check whether they're genuine solutions of (3.6) or spurious artifacts of squaring, before concluding there's an actual crossing.

This asymmetry is why the lemma is phrased as "free of arbitrage **if**" rather than "if and only if."

get a full surface, in practice what people do (and what the paper assumes) is: fit a *separate* raw SVI curve to the market data at each observed expiry. So if your data has expiries $t_1, t_2, \dots, t_n$ (say, 1 week, 1 month, 3 months, ...), you end up with n *independent* parameter sets:

$$\chi_1 = \{a_1, b_1, \rho_1, m_1, \sigma_1\} \text{ for } t_1, \quad \chi_2 = \{a_2, b_2, \rho_2, m_2, \sigma_2\} \text{ for } t_2, \quad \dots$$

Stack these curves side by side (one per maturity) and that collection *is* your "surface" — but it's a **discrete** stack of curves, not a single smooth 2-variable formula. There's no built-in rule connecting χ_1 to χ_2 ; they were fit independently.

Remark 3.2. *By a careful study of the min and $w(\cdot, t_2)$, it is possible to determine a set calendar spread arbitrage. However these con parameters and will hence not match the con*

FOR ALL k use Polynomial

The direct comparison — " $w(k, t_1) \leq w(k, t_2)$ for all k " — is the *true definition* of what you want. But you can't literally test "for all $k \in \mathbb{R}$ " by hand or by brute-force evaluation, because $\mathbb{R}$ is infinite. You need some way to know whether the difference

$$f(k) := w(k, t_2) - w(k, t_1)$$

ever changes sign.

Solving the quartic is precisely the tool that answers that question exactly. Since $f(k) = 0$ leads (after algebraic manipulation) to the quartic (3.7), finding all real roots of that quartic tells you all the points where the two curves *could* touch or cross. That's not a different check — it's the exact, closed-form way to determine whether the "for all k " comparison holds.

Example 3.1. (From Axel Vogt on wilmott.com)

$$(a, b, m, \rho, \sigma) = (-0.0410, 0.1331, 0.3586, 0.3060, 0.4153), \quad (3.8)$$

with $t = 1$. These parameters give rise to the total variance smile w and the function g (defined in (2.1)) on Figure 1, where the negative density is clearly visible.

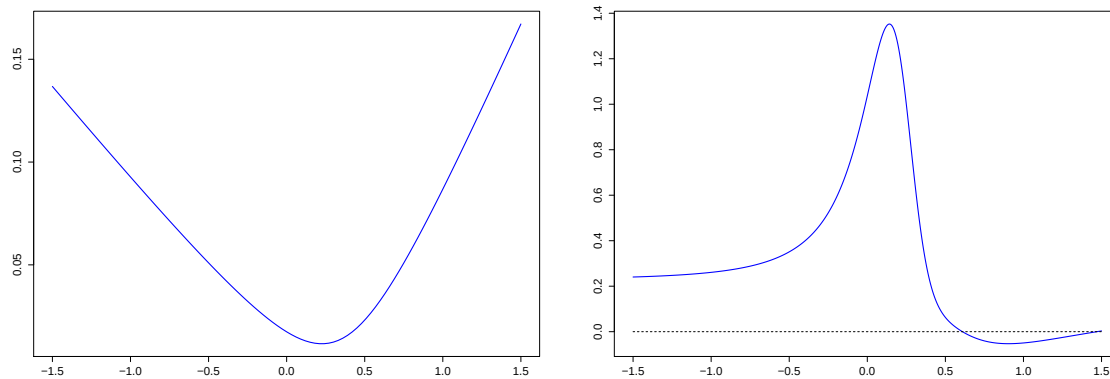


Figure 1: Plots of the total variance smile w (left) and the function g defined in (2.1) (right), using the parameters (3.8).

This produces a smile $w(k)$ (Figure 1, left) that *looks* perfectly reasonable — smooth, convex, well-behaved to the eye. But plotting $g(k)$ (Figure 1, right) reveals it dips **below zero** over a range of strikes. By Lemma 2.2, this means the implied risk-neutral density is negative there — the smile is arbitrageable, despite passing a naive visual/convexity check.

4 Surface SVI: A surface free of static arbitrage

We now introduce a class of SVI volatility surfaces—which we shall call SSVI (for ‘Surface SVI’)—as an extension of the natural parameterization (3.2). For any maturity $t \geq 0$,

So the paper's point is: in this discrete setting, the *meaning* of " w is non-decreasing in t " degrades to something you *can* check — namely, comparing each pair of adjacent curves directly:

$$w(k, t_1) \leq w(k, t_2) \quad \text{for all } k,$$

$$w(k, t_2) \leq w(k, t_3) \quad \text{for all } k,$$

and so on. If curve 2 lies entirely above (or touching) curve 1 everywhere, and curve 3 lies entirely above curve 2 everywhere, etc., then effectively your discrete "staircase" of curves is monotonically increasing in t — which is the discrete analogue of $\partial_t w \geq 0$.

Picture

Imagine plotting $w(k, t_1)$ and $w(k, t_2)$ on the same (k, w) axes, as two curves:

- **Good case (no arbitrage):** the t_2 curve sits completely above the t_1 curve, for every k . They never touch or cross.
- **Bad case (calendar arbitrage):** the two curves cross somewhere — at some log-strike k^* , curve t_1 is actually *higher* than curve t_2 . That crossing point is where an arbitrage trade exists (sell the short-dated option, buy the long-dated one, at that strike).

Solve for the roots — don't just eyeball/scan. Here's the practical procedure:

1. **Compute the quartic coefficients** from χ_1, χ_2 (given explicitly, if tediously, in terms of the ten raw SVI parameters).
2. **Find its real roots** (closed-form via Cardano/Ferrari, or just numerically — e.g. `numpy.roots`).
3. **Case A — no real roots:** $f(k)$ never crosses zero, so it has a fixed sign everywhere. Just check the sign at one convenient point (e.g. $k = 0$) to see whether $w(\cdot, t_2) \geq w(\cdot, t_1)$. If yes, calendar-arbitrage-free between this pair — done.
4. **Case B — real roots exist:** these are only *candidate* crossing points, because the quartic came from squaring equation (3.6) twice, which can introduce spurious roots that don't actually solve the original equation. So for each real root k^* , plug it back into the **original** unsquared equation (3.6) and check it's actually satisfied (up to numerical tolerance). Discard any that aren't genuine.
5. For the genuine roots that remain, check the sign of $f(k)$ in between and around them — if f changes sign anywhere, the curves genuinely cross, and you have calendar arbitrage between t_1, t_2 at that strike.

$$\text{implied total var: } w(k, t) = \sigma_{\text{BS}}^2(k, t) \times t$$

define the at-the-money (ATM) implied total variance $\theta_t := \sigma_{\text{BS}}^2(0, t)t$. We shall assume that the function θ is at least of class $\mathcal{C}^1$ on $\mathbb{R}_+^*$. An ATM option with zero time to expiry has no value so $\theta_0 := \lim_{t \rightarrow 0} \theta_t = 0$. Instead of fitting five independent parameters *per maturity*, SSVI expresses the whole surface as one function of (k, θ_t) :

Definition 4.1. Let φ be a smooth function from $\mathbb{R}_+^*$ to $\mathbb{R}_+^*$ such that the limit $\lim_{t \rightarrow 0} \theta_t \varphi(\theta_t)$ exists in $\mathbb{R}$. We refer to as SSVI the surface defined by

$$w(k, \theta_t) = \frac{\theta_t}{2} \left\{ 1 + \rho \varphi(\theta_t) k + \sqrt{(\varphi(\theta_t) k + \rho)^2 + (1 - \rho^2)} \right\}. \quad (4.1)$$

From Section 3, SSVI corresponds to the natural SVI volatility surface parameterization (3.2) with $\chi_N = \{0, 0, \rho, \theta_t, \varphi(\theta_t)\}$. Note that this representation amounts to considering the volatility surface in terms of ATM variance time, instead of standard calendar time, similar in spirit to the stochastic subordination of [7].

This is exactly natural SVI (3.2) with $\chi_N = \{0, 0, \rho, \theta_t, \varphi(\theta_t)\}$ — i.e. $\Delta = 0, \mu = 0, \omega = \theta_t, \zeta = \varphi(\theta_t)$. The whole surface now only requires **two ingredients**: a correlation parameter $\rho \in (-1, 1)$ (constant across all maturities), and a **shape function** $\varphi : \mathbb{R}_+^* \rightarrow \mathbb{R}_+^*$ that controls how the smile's steepness scales with the ATM variance level θ_t . This is the key structural move: instead of five free parameters per slice, you get essentially two free objects — ρ and φ — for the *entire* surface.

Why this is smart. θ_t is directly observable in the market (it's just the ATM implied volatility squared times t), so a trader always knows it, even for maturities that aren't explicitly quoted. Using θ_t instead of calendar time t as the "clock" is described as analogous to stochastic time-change / subordination (Clark [7]) — you're re-parameterizing the surface in terms of **volatility-time** rather than calendar time.

in classical models which may shed some light on how to impose dynamics on SSVI.

The ATM implied total variance is $\theta_t = \sigma_{\text{BS}}^2(0, t)t$ and the ATM volatility skew is given by

$$\frac{\partial_k \sigma_{\text{BS}}(k, t)|_{k=0}}{2\sqrt{\theta_t t}} = \frac{1}{2\sqrt{\theta_t t}} \frac{\partial_k w(k, \theta_t)|_{k=0}}{\theta_t} = \frac{\rho \sqrt{\theta_t}}{2\sqrt{t}} \varphi(\theta_t). \quad (4.2)$$

Furthermore the smile is symmetric around at-the-money if $w(k, \theta_t) = w(-k, \theta_t)$ holds iff $\rho = 0$, which is consistent with [4, Theorem 3.4] which states that in a standard stochastic volatility model, the smile is symmetric if and only if the correlation between the stock price and its instantaneous volatility is null. Since $\theta_0 = 0$, we have at time $t = 0$:

$$w(k, \theta_0) = \frac{1}{2} \phi_0 (\rho k + |k|), \quad \text{for any } k \in \mathbb{R}, \quad (4.3)$$

where $\phi_0 := \lim_{\theta \rightarrow 0} \theta \varphi(\theta)$. If $\phi_0 = 0$ is characteristic of stochastic volatility models as in Example 4.1; $\phi_0 > 0$ as in Example 4.2 gives a V-shaped time zero smile which is characteristic of models with jumps and in particular, characteristic of empirically observed volatility surfaces. For notational convenience, we shall always assume that $\lim_{t \nearrow \infty} \theta_t = \infty$. As proved in [24], this is equivalent (assuming no interest rate) to the stock price (assumed to be a non-negative martingale) converging to zero as t tends to infinity. Although this holds in many popular models (Black-Scholes, Heston, exponential Lévy), this is not always true, see [19] for counter-examples. If $\lim_{t \nearrow \infty} \theta_t$ is finite, all our results remain valid, but only on the support of the function $t \mapsto \theta_t$.

The following theorem gives precise necessary and sufficient conditions to ensure that the SSVI volatility surface (4.1) is free of calendar spread arbitrage (Lemma 2.1) and also matches the term structure of ATM volatility and the term structure of the ATM volatility skew.

Since $\partial_t w(k, \theta_t) = \partial_\theta w(k, \theta_t) \cdot \partial_t \theta_t$, calendar-arbitrage-freedom ($\partial_t w \geq 0$) reduces to two separate, decoupled conditions:

Theorem 4.1. The SSVI surface (4.1) is free of calendar spread arbitrage if and only if

1. $\partial_t \theta_t \geq 0$, for all $t \geq 0$;
2. $0 \leq \partial_\theta(\theta \varphi(\theta)) \leq \frac{1}{\rho^2} \left(1 + \sqrt{1 - \rho^2}\right) \varphi(\theta)$, for all $\theta > 0$,

where the upper bound is infinite when $\rho = 0$.

In particular, this theorem implies that the SSVI surface (4.1) is free of calendar spread arbitrage if the skew in total variance terms is monotonically increasing in trading time and the skew in implied variance terms is monotonically decreasing in trading time. In practice, any reasonable skew term structure that a trader defines has these properties.

Proof. Since the definition of calendar spread arbitrage does not depend on the log-moneyness k , there is no loss of generality in assuming k fixed. First note that $\partial_t w(k, \theta_t) = \partial_\theta w(k, \theta_t) \partial_t \theta_t$ so the SSVI volatility surface (4.1) is free of calendar spread arbitrage if $\partial_\theta w(k, \theta) \geq 0$ for all $\theta > 0$.

Consider first the case $|\rho| < 1$. To proceed, we compute, for any $\theta > 0$,

$$2\partial_\theta w(k, \theta) = \psi_0(x, \rho) + \gamma(\theta)\psi_1(x, \rho),$$

with $x := k\varphi(\theta)$, $\gamma(\theta) := \partial_\theta(\theta\varphi(\theta))/\varphi(\theta)$,

$$\psi_0(x, \rho) := 1 + \frac{1 + \rho x}{\sqrt{x^2 + 2\rho x + 1}} \quad \text{and} \quad \psi_1(x, \rho) := x \left\{ \frac{x + \rho}{\sqrt{x^2 + 2\rho x + 1}} + \rho \right\}.$$

For any $|\rho| < 1$, $\psi_0(x, \rho)$ is strictly positive for all $x \in \mathbb{R}$. Now define the set

$$\mathcal{D}_\rho = \begin{cases} (-\infty, 0) \cup (-2\rho, \infty), & \text{if } \rho < 0, \\ (-\infty, -2\rho) \cup (0, \infty), & \text{if } \rho > 0, \\ \mathbb{R} \setminus \{0\}, & \text{if } \rho = 0. \end{cases}$$

Then $\psi_1(\cdot, \rho) > 0$ if $x \in \mathcal{D}_\rho$ and $\psi_1(\cdot, \rho) < 0$ if $x \in \mathbb{R} \setminus (\mathcal{D}_\rho \cup \{0, -2\rho\})$. It follows that

$$\partial_\theta w(k, \theta) \geq 0 \text{ if and only if } \begin{cases} \gamma(\theta) \geq -\frac{\psi_0(x, \rho)}{\psi_1(x, \rho)}, & \text{for } x \in \mathcal{D}_\rho, \\ \gamma(\theta) \leq -\frac{\psi_0(x, \rho)}{\psi_1(x, \rho)}, & \text{for } x \in \mathbb{R} \setminus (\mathcal{D}_\rho \cup \{0, -2\rho\}), \end{cases} \quad (4.4)$$

When $x \in \{0, -2\rho\}$, then $\psi_1(x, \rho) = 0$ and so $\partial_\theta w(k, \theta) \geq 0$. The inequalities (4.4) thus give necessary and sufficient conditions for absence of calendar spread arbitrage for any given $x \in \mathbb{R}$. To determine the tightest possible bounds on $\gamma(\theta)$, we compute

$$\sup_{x \in \mathcal{D}_\rho} -\frac{\psi_0(x, \rho)}{\psi_1(x, \rho)} = 0 \quad \text{and} \quad \inf_{x \in \mathbb{R} \setminus (\mathcal{D}_\rho \cup \{0, -2\rho\})} -\frac{\psi_0(x, \rho)}{\psi_1(x, \rho)} = \frac{1 + \sqrt{1 - \rho^2}}{\rho^2}.$$

The supremum in the first equality is never attained (the function increases to zero from below as $|x|$ tends to infinity). However the infimum in the second equality is attained at $x = -\rho \notin \mathcal{D}_\rho$. It follows that

$$\partial_\theta w(k, \theta) \geq 0 \text{ if and only if } 0 \leq \gamma(\theta) \leq \frac{1 + \sqrt{1 - \rho^2}}{\rho^2}.$$

Note that when $\rho = 0$, the infimum above is taken over an empty set, and there is hence no upper bound.

When $\rho = 1$, for any $(k, \theta) \in \mathbb{R} \times (0, \infty)$, we have

$$\partial_\theta w(k, \theta) = \left(1 + \frac{1+x}{\sqrt{(1+x)^2}}\right) (1 + \gamma(\theta)x) = \begin{cases} 2(1 + \gamma(\theta)x) & \text{if } x \geq -1, \\ 0 & \text{otherwise.} \end{cases}$$

Obviously, $\partial_\theta w(k, \theta) \geq 0$ if $x \geq 0$. For $x > -1$, clearly $\partial_\theta w(k, \theta) \geq 0$ if and only if $\gamma(\theta) \in [0, 1]$. Similarly, with $\rho = -1$, we have

$$\partial_\theta w(k, \theta) = \left(1 + \frac{1-x}{\sqrt{(1-x)^2}}\right) (1 - \gamma(\theta)x) = \begin{cases} 2(1 - \gamma(\theta)x) & \text{if } x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Again $\partial_\theta w(k, \theta) \geq 0$ if $x \leq 0$, and for $x \leq 1$, $\partial_\theta w(k, \theta) \geq 0$ if and only if $\gamma(\theta) \in [0, 1]$. $\square$

The following lemma is a straightforward consequence of (3.3) and (3.5).

Lemma 4.1. *The SVI-JW parameters associated with the SSVI surface (4.1) are*

$$\begin{aligned} v_t &= \theta_t/t, \\ \psi_t &= \frac{1}{2} \rho \sqrt{\theta_t} \varphi(\theta_t), \\ p_t &= \frac{1}{2} \sqrt{\theta_t} \varphi(\theta_t) (1 - \rho), \\ c_t &= \frac{1}{2} \sqrt{\theta_t} \varphi(\theta_t) (1 + \rho), \\ \tilde{v}_t &= \frac{\theta_t}{t} (1 - \rho^2). \end{aligned}$$

We now give several examples of SSVI implied volatility surfaces (4.1).

Example 4.1. A Heston-like parameterization

Consider the function φ defined by

$$\varphi(\theta) \equiv \frac{1}{\lambda\theta} \left\{ 1 - \frac{1 - e^{-\lambda\theta}}{\lambda\theta} \right\},$$

with $\lambda > 0$. Then for all $\theta > 0$, we immediately obtain

$$\partial_\theta (\theta\varphi(\theta)) = \frac{e^{-\lambda\theta} (e^{\lambda\theta} - 1 - \lambda\theta)}{\lambda^2\theta^2} > 0 \quad \text{and} \quad \frac{\partial_\theta (\theta\varphi(\theta))}{\varphi(\theta)} = \frac{1 - (1 + \lambda\theta)e^{-\lambda\theta}}{e^{-\lambda\theta} + \lambda\theta - 1}.$$

For any $\lambda > 0$, the map $\theta \mapsto \partial_\theta (\theta \varphi(\theta)) / \varphi(\theta)$ is strictly decreasing on $(0, \infty)$ with limit as θ tends to zero equal to one. Since the quantity $(1 + \sqrt{1 - \rho^2}) / \rho^2$ is greater than one for any $\rho \in [-1, 1]$, the conditions of Theorem 4.1 are satisfied. This function is consistent with the implied variance skew in the Heston model as shown in [14, Equation 3.19].

Example 4.2. Power-law parameterization

Consider $\varphi(\theta) = \eta \theta^{-\gamma}$ with $\eta > 0$ and $0 < \gamma < 1$. Then $\partial_\theta (\theta \varphi(\theta)) / \varphi(\theta) = 1 - \gamma \in (0, 1)$ holds for all $\theta > 0$, and hence the conditions of Theorem 4.1 are satisfied. In particular if $\gamma = 1/2$ then Lemma 4.1 implies that the SVI-JW parameters ψ_t , p_t , and c_t associated with the SSVI volatility surface (4.1) are constant and independent of the time to expiration t . Furthermore, Equation 4.2 implies that the ATM volatility skew is given by

$$\partial_k \sigma_{BS}(k, t)|_{k=0} = \frac{\rho \eta}{2\sqrt{t}}.$$

The following theorem provides sufficient conditions for a SSVI surface (4.1) to be free of butterfly arbitrage.

Theorem 4.2. The SSVI volatility surface (4.1) is free of butterfly arbitrage if the following conditions are satisfied for all $\theta > 0$:

1. $\theta \varphi(\theta) (1 + |\rho|) < 4$;
2. $\theta \varphi(\theta)^2 (1 + |\rho|) \leq 4$.

This is the actual **result** — the thing you check when calibrating. It's phrased directly in terms of the SSVI parameterization's native objects: θ (ATM total variance) and $\varphi(\theta)$ (the shape function), plus ρ .

Proof. For ease of notation, we suppress the explicit dependence of θ and φ on t . By symmetry, it is enough to prove the theorem for $0 \leq \rho < 1$. We shall therefore assume so, and we define $z := \varphi k$. The function g defined in (2.1) reads

$$g(z) = \frac{f(z)}{64 (z^2 + 2z\rho + 1)^{3/2}},$$

where

$$f(z) := a - b\varphi^2\theta - \frac{c}{16}\varphi^2\theta^2,$$

and where a , b and c depend on z . In the following, we frequently use the inequality

$$z^2 + 2z\rho + 1 = (z + \rho)^2 + 1 - \rho^2 \geq 0.$$

Computing the coefficient of $\varphi^2\theta^2$ in $f(z)$ explicitly gives

$$\begin{aligned} c &= \sqrt{z^2 + 2z\rho + 1} \left\{ (1 + \rho^2) (z + \rho)^2 + 2\rho(z + \rho)\sqrt{z^2 + 2z\rho + 1} + (1 - \rho^2) \rho^2 \right\} \\ &\geq \sqrt{z^2 + 2z\rho + 1} \left\{ (1 + \rho^2) (z + \rho)^2 + 2\rho(z + \rho)^2 + (1 - \rho^2) \rho^2 \right\} \\ &= \sqrt{z^2 + 2z\rho + 1} \left\{ (1 + \rho)^2 (z + \rho)^2 + (1 - \rho^2) \rho^2 \right\} \geq 0. \end{aligned}$$

Thus if

$$0 \leq \theta\varphi \leq \frac{4}{1+\rho} \quad \text{and} \quad 0 \leq \theta\varphi^2 \leq \frac{4}{1+\rho},$$

we have

$$f(z) \geq \begin{cases} a - \frac{4b}{1+\rho} - \frac{c}{(1+\rho)^2} =: f_1(z), & \text{if } b \geq 0, \\ a - \frac{c}{(1+\rho)^2} =: f_2(z), & \text{if } b < 0. \end{cases}$$

It is then straightforward to verify that

$$\begin{aligned} \frac{2f_1(z)}{(1+\rho)^2} &= \sqrt{z^2 + 2z\rho + 1} \{z^2\rho - z(1-\rho)\rho + 2(1+\rho)(1-\rho^2) + \rho\} \\ &\quad + \rho(z+\rho)^2 + 3\rho(1-\rho^2) + 2(1-\rho^2) - z\rho(z^2 + 2z\rho + 1), \end{aligned}$$

which is clearly positive for $z < 0$. To see that $f_1(z)$ is also positive when $z > 0$, we rewrite it as

$$\begin{aligned} &\frac{2f_1(z)}{(1+\rho)^2} \\ &= \left\{ \sqrt{z^2 + 2z\rho + 1} - (z + \rho) \right\} \left\{ \rho \left(z - \frac{1-\rho}{2} \right)^2 + 2(1+\rho)(1-\rho^2) + \rho \left(1 - \frac{(1-\rho)^2}{4} \right) \right\} \\ &\quad + (1+\rho) \{ z(2-\rho^2) + 2(1+\rho)(1-\rho^2) + \rho \}. \end{aligned}$$

Consider now the function $f_2(z)$. It is straightforward to verify that

$$f_2(z) = -\frac{2z^3\rho}{(1+\rho)^2} + (z^2 + 2z\rho + 1)^{3/2} + 2(z^2 + 2z\rho + 1) + \sqrt{z^2 + 2z\rho + 1}$$

which is positive by inspection if $z < 0$. To see that $f_2(z)$ is also positive when $z > 0$, we rewrite it as

$$\begin{aligned} f_2(z) &= z^3 \frac{1+\rho^2}{(1+\rho)^2} + 3z^2\rho + 2(z^2 + 2z\rho + 1) \\ &\quad + (z^2 + 2z\rho + 1) \left\{ \sqrt{z^2 + 2z\rho + 1} - (z + \rho) \right\} \\ &\quad + \sqrt{z^2 + 2z\rho + 1} + 2z\rho^2 + z + \rho. \end{aligned}$$

Thus $f(z) \geq 0$ in all cases. From Lemma 2.2, we are left to prove that $\lim_{k \rightarrow \infty} d_+(k) = -\infty$. A straightforward computation shows that this is satisfied as soon as Condition 1 in Theorem 4.2 holds. $\square$

Remark 4.2. A SSVI volatility surface (4.1) is free of butterfly arbitrage if

$$\sqrt{v_t t} \max(p_t, c_t) < 2, \quad \text{and} \quad (p_t + c_t) \max(p_t, c_t) \leq 2,$$

hold for all $t > 0$. The proof follows from Lemma 4.1 by re-expressing Conditions 1 and 2 of Theorem 4.2 in terms of SVI-JW parameters.

the same conditions in THM 4.2, rewritten in SVI-JW language

Sufficient vs. necessary — the core distinction

When a theorem says "condition X implies no arbitrage," that's a **sufficient** condition: having X is *enough* to guarantee the outcome, but there might be other ways to get the same outcome without X .

- $X \Rightarrow$ no arbitrage (sufficient)

This leaves a real question open: is X also **necessary** — meaning if you *don't* have X , are you *guaranteed* to have arbitrage? Or could there be some other, non- X way to still be fine?

- no arbitrage $\Rightarrow X$ (necessary — the reverse arrow)

If both directions hold, X is "if and only if" — the exact boundary of good behavior, not just a safe zone within it.

Theorem 4.2 proved: if $\theta\varphi(\theta)(1 + |\rho|) < 4$ (strictly less than 4) for all θ , then no butterfly arbitrage. That's the " $X \Rightarrow$ good" direction — sufficiency.

What it *didn't* tell you: what happens if $\theta\varphi(\theta)(1 + |\rho|)$ creeps up to exactly 4, or above 4? Maybe you're still fine — maybe the " < 4 " was just a conservative, extra-safe threshold the proof happened to produce, and the *real* danger zone starts somewhere further out, like 5 or 10. Without more information, you wouldn't know whether 4 is the actual edge of the cliff or just a comfortable margin before it.

Lemma 4.2 answers this by proving the *other* direction: if the surface is arbitrage-free, then necessarily $\theta\varphi(\theta)(1 + |\rho|) \leq 4$. In other words: **go above 4, and you are guaranteed to have arbitrage** — there is no clever exception, no lucky combination of other parameters that saves you once you cross that line. That confirms 4 really is the cliff edge, not just a conservative buffer.

The following lemma shows that **Theorem 4.2 is almost if-and-only-if**, how tight Theorem 4.2 actually is

Lemma 4.2. *The SSVI volatility surface (4.1) is free of butterfly arbitrage only if*

$$\theta\varphi(\theta)(1 + |\rho|) \leq 4, \quad \text{for all } \theta > 0.$$

X is sufficient for Y (having X guarantees Y).
 Y is necessary for X (X cannot happen without Y).

Moreover *if $\theta\varphi(\theta)(1 + |\rho|) = 4$, the SSVI surface is free of butterfly arbitrage only if*

$$\theta\varphi(\theta)^2(1 + |\rho|) \leq 4.$$

Whenever you evaluate " $X \Rightarrow Y$ " (X implies Y):

- The left side (X) is sufficient for the right side (Y).
- The right side (Y) is necessary for the left side (X).

Thus *Condition 1 of Theorem 4.2 is necessary and Condition 2 is tight*.

Why the strict/non-strict distinction matters (" < 4 " vs " ≤ 4 ")

- Theorem 4.2 guarantees safety only for values **strictly less than 4**.
- Lemma 4.2 shows arbitrage-freedom requires values **at most 4** (i.e., $= 4$ is still theoretically permitted, not automatically bad).

So the single point $\theta\varphi(\theta)(1 + |\rho|) = 4$ exactly is a knife-edge: Theorem 4.2 doesn't promise it's safe (its condition demanded *strictly* less), and Lemma 4.2 doesn't prove it's automatically arbitrage — it's a borderline case that could go either way depending on other details. That's why the paper calls Theorem 4.2's bound "essentially as tight as it can be" — there's no room to loosen " < 4 " to something like " < 5 "; the true boundary sits right at 4, and Theorem 4.2 already reaches all the way up to it (short of the single boundary point itself).

THE RESULT FOLLOWS BY INSPECTION.

What "condition 2 is tight exactly at the boundary" means

This is a more refined statement about that exact knife-edge case ($= 4$). Lemma 4.2 shows: **if you're sitting exactly on the boundary ($\theta\varphi(\theta)(1 + |\rho|) = 4$), then you additionally need condition 2 ($\theta\varphi(\theta)^2(1 + |\rho|) \leq 4$) to still be arbitrage-free** — condition 2 isn't optional there, it becomes required.

Away from that exact boundary (i.e., strictly below 4), Theorem 4.2 already told you condition 2 alongside condition 1 is enough — but Lemma 4.2 is telling you something extra: right at the edge, condition 2 stops being a "nice to also have" add-on and becomes load-bearing. If you dropped condition 2 while sitting exactly at condition 1's boundary, you could genuinely have arbitrage.

Remark 4.3. *The asymptotic behavior of SSVI (4.1) as $|k|$ tends to infinity is*

$$w(k, \theta_t) = \frac{(1 \pm \rho)\theta_t}{2}\varphi(\theta_t)|k| + \mathcal{O}(1), \quad \text{for any } t > 0.$$

We thus observe that the condition $\theta\varphi(\theta)(1 + |\rho|) \leq 4$ of Theorem 4.2 corresponds to the upper bound of 2 on the asymptotic slope established by Lee [23] and so again, Condition 1 of Theorem 4.2 is necessary.

The following corollary follows directly from Theorems 4.1 and 4.2.

Corollary 4.1. *The SSVI surface (4.1) is free of static arbitrage if the following conditions are satisfied:*

1. $\partial_t\theta_t \geq 0$, for all $t > 0$
2. $0 \leq \partial_\theta(\theta\varphi(\theta)) \leq \frac{1}{\rho^2} \left(1 + \sqrt{1 - \rho^2}\right) \varphi(\theta)$, for all $\theta > 0$;
3. $\theta\varphi(\theta)(1 + |\rho|) < 4$, for all $\theta > 0$;
4. $\theta\varphi(\theta)^2(1 + |\rho|) \leq 4$, for all $\theta > 0$.

Remark 4.4. *Consider the function $\varphi(\theta) = \eta\theta^{-\gamma}$ with $\eta > 0$ from Example 4.2, then Condition 2 imposes $\gamma \in (0, 1)$. From Condition 3, such surfaces can be free of static arbitrage only up to some maximum expiry. Take for instance the simple case $\theta_t := \sigma^2 t$ for some $\sigma > 0$. Then the map $\psi : t \mapsto \theta_t\varphi(\theta_t)(1 + |\rho|) - 4$ is clearly strictly increasing with $\psi(0) = -4$ and $\lim_{t \rightarrow \infty} \psi(t) = \infty$. Therefore there exists $t_0^* > 0$ such that $\psi(t) \leq 0$ for $t \leq t_0^*$. The map $\psi_2 : t \mapsto \theta_t\varphi(\theta_t)^2(1 + |\rho|) - 4$ is*

- strictly increasing if $\gamma \in (0, 1/2)$ with $\psi_2(0) = -4$ and $\lim_{t \rightarrow \infty} \psi(t) = +\infty$; there exists $t_1^* > 0$ such that $\psi_2(t) \leq 0$ for $t \leq t_1^*$.
- strictly decreasing if $\gamma \in (1/2, 1)$ with $\lim_{t \rightarrow 0} \psi_2(0) = +\infty$ and $\lim_{t \rightarrow \infty} \psi(t) = -4$; there exists $t_1^* > 0$ such that $\psi_2(t) \leq 0$ for $t \geq t_1^*$.
- constant if $\alpha = 1/2$ with $\psi_2 \equiv -4$.

When $\gamma \in (0, 1/2)$, the surface is guaranteed to be free of static arbitrage only for $t \leq t_0^* \wedge t_1^*$. For $\gamma \in (1/2, 1)$, this remains true only for $t \in (0, t_0^*) \cap (t_1^*, \infty)$ (which may be empty). When $\gamma = 1/2$, static arbitrage cannot occur for $t \leq t_0^*$. However, the behavior for large θ can be easily modified so as to ensure that the entire surface is free of static arbitrage. For example, the choice

$$\varphi(\theta) = \frac{\eta}{\theta^\gamma (1 + \theta)^{1-\gamma}} \quad (4.5)$$

gives a surface that is completely free of static arbitrage provided that $\eta(1 + |\rho|) \leq 2$.

Remark 4.5. In the Heston-like parameterization of Example 4.1, note that

$$\lim_{\theta \rightarrow +\infty} \theta \varphi(\theta) (1 + |\rho|) = \frac{1 + |\rho|}{\lambda}.$$

Therefore Condition 3 of Corollary 4.1 imposes $\lambda \geq (1 + |\rho|)/4$.

You've built an SSVI surface (Definition 4.1) that's fully arbitrage-free — it satisfies Corollary 4.1 (calendar-clean, per Theorem 4.1, and butterfly-clean, per Theorem 4.2). Good. But this surface is built from just **two** shared objects: a single correlation ρ , and a single shape function $\varphi(\theta)$, applied identically across every maturity.

That's very restrictive. With only two "global" objects to tune, you can match maybe 2-3 features of one particular smile really well (its level, its skew, maybe curvature) — but you don't have enough free parameters left to also perfectly match every feature of every other maturity's smile. Something has to give.

The question Theorem 4.3 answers

Can I add something to the surface, per-maturity, to fix this up — without breaking the arbitrage-free guarantee I already worked to establish?

independent theorem provides a way to ensure the surface is guaranteed to be free of static arbitrage by

The statement, in plain terms

Yes — you're allowed to add a "time-only" bump to the whole surface:

$$w_\alpha(k, \theta_t) = w(k, \theta_t) + \alpha_t$$

Theorem 4.3. Let $(k, t) \mapsto w(k, t)$ be a SSVI volatility surface (4.1) satisfying the conditions of Corollary 4.1 (in particular free of static arbitrage), and $\alpha : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ a non-negative and increasing function of time. Then the volatility surface $(k, t) \mapsto w_\alpha(k, \theta_t) := w(k, \theta_t) + \alpha_t$ is also free of static arbitrage.

It's a **calibration flexibility tool**. Once your global ρ , φ are locked in (fit to nail level and skew everywhere), α_t gives you one more free number per maturity to tune — e.g., to push a given slice's level up slightly to better match an additional observed market quantity — all without invalidating the arbitrage-free guarantee you already have. It's the mechanism that lets you loosen SSVI's rigidity just enough to get closer fits, while staying inside the provably safe region.

Proof. Since $\partial_t w_\alpha(k, \theta_t) := \partial_t w(k, \theta_t) + \partial_t \alpha_t$, spread arbitrage if $\partial_t \alpha_t \geq 0$ and $\alpha_t \geq 0$. V we shall use the simplified notation $w(k) := w(k, \theta_t)$, and likewise $w_\alpha(k) := w_\alpha(k, \theta_t)$. Similarly, in view of (2.1), we shall define the map $g_\alpha(k)$, where the function w is replaced by w_α . We consider the case $\rho < 0$ since the case $\rho > 0$ follows by symmetry, and the result is obvious when $\rho = 0$. Let us consider the function $G_\alpha : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$G_\alpha(k) := g(k) - g_\alpha(k), \quad \text{for all } k \in \mathbb{R},$$

and let $k^* := -2\rho/\varphi(\theta_t) > 0$ be the unique solution to the equation $w'(k) = 0$. We can compute explicitly the following:

$$G_\alpha(k) = \frac{w'(k)}{4} \left(\frac{1}{w_\alpha(k)} - \frac{1}{w(k)} \right) \left(4k + w'(k) - w'(k)k^2 \left(\frac{1}{w_\alpha(k)} + \frac{1}{w(k)} \right) \right),$$

which implies

$$\partial_\alpha G_\alpha(k) = -\frac{w'(k)}{4} \frac{(w'(k) + 4k) w_\alpha(k) - 2k^2 w'(k)}{w_\alpha(k)^3}. \quad (4.6)$$

Since $w'(0) = \rho\theta_t\varphi(\theta_t) < 0$ the equation $w'(k) + 4k = 0$ has a unique solution $k_* > 0$, and $w'(k) + 4k$ is strictly positive for any $k > k_*$ and strictly negative when $k < k_*$. By strict convexity of the function w it also follows that $k_* < k^*$. Therefore for any $k \in (k_*, k^*)$, the two inequalities $w'(k) < 0$ and $w'(k) + 4k > 0$ hold, and therefore $\partial_\alpha G_\alpha(k) > 0$. Since by construction $G_0(k) = 0$, we therefore conclude that $g(k) > g_\alpha(k)$ for any $k \in (k_*, k^*)$. For $k \notin (k_*, k^*)$, the inequality $g(k) < g_\alpha(k)$ holds as soon as $\partial_\alpha G_\alpha(k) < 0$. Consider first the case $k > k^*$. We can rewrite (4.6) as

$$\partial_\alpha G_\alpha(k) = -\frac{w'(k)}{4} \frac{2k [2w_\alpha(k) - kw'(k)] + w_\alpha(k)w'(k)}{w_\alpha(k)^3}.$$

so that it suffices to prove the inequality $2w_\alpha(k) - kw'(k) > 0$ for any $k > k^*$. It suffices to prove $\partial_\alpha G_\alpha(k) < 0$ for then we have the inequality $g_\alpha(k) > g(k) \geq 0$ and there is no butterfly arbitrage.

First consider the case $k > k^*$, so that $w'(k) > 0$. Recall that a continuously differentiable function f is convex on the interval (a, b) if and only if $f(x) - f(y) \geq f'(x)(x - y)$ for all $(x, y) \in (a, b)$. Setting $x = k$ and $y = 0$, we conclude that $2w_\alpha(k) - kw'(k) > 0$ since $w_\alpha(0) \geq 0$. It follows that $\partial_\alpha G_\alpha(k) < 0$ for any $k > k^*$.

For any $k < 0$, we always have $w'(k) < 0$, the inequality $2w_\alpha(k) - kw'(k) > 0$ follows by convexity as above, and hence $\partial_\alpha G_\alpha(k) < 0$ for any $k < 0$. We prove here that $g_\alpha(k) \geq g_\alpha(0)$ for all such k . Since we already showed that $g_\alpha(0) > 0$, the result follows. From the definition of g_α and (2.1),

$$\begin{aligned} g_\alpha(k) - g_\alpha(0) &= \left(1 - \frac{kw'(k)}{2(w(k) + \alpha)}\right)^2 - 1 \\ &\quad - \frac{w'(k)^2}{4} \left(\frac{1}{w(k) + \alpha} + \frac{1}{4}\right) + \frac{w'(0)^2}{4} \left(\frac{1}{w(0) + \alpha} + \frac{1}{4}\right) \\ &\quad + \frac{w''(k)}{2} - \frac{w''(0)}{2}. \end{aligned} \quad (4.7)$$

A straightforward analysis shows that the function $k \mapsto w''(k)$ is strictly increasing on the interval $(0, k^*/2)$ and strictly decreasing on $(k^*/2, k^*)$. The easy computation $w''(0) = w''(k^*)$ implies that $w''(k) \geq w''(0)$ on $(0, k^*)$. Also, $w'(0)^2 > w'(k)^2$ on $(0, k^*)$. Simplifying (4.7), it follows that

$$\begin{aligned} g_\alpha(k) - g_\alpha(0) &\geq \left(1 - \frac{kw'(k)}{2(w(k) + \alpha)}\right)^2 - 1 + \frac{1}{4} \left(\frac{w'(0)^2}{w(0) + \alpha} - \frac{w'(k)^2}{w(k) + \alpha}\right) \\ &\geq \frac{1}{4} \left(\frac{w'(0)^2}{w(0) + \alpha} - \frac{w'(k)^2}{w(k) + \alpha}\right) - \frac{kw'(k)}{w(k) + \alpha}. \end{aligned}$$

Note that $w'(k)^2 \leq w'(0) w'(k) \leq w'(0)^2$ on the interval $(0, k^*)$ so

$$g_\alpha(k) - g_\alpha(0) \geq \frac{w'(0) w'(k)}{4} \left(\frac{1}{w(0) + \alpha} - \frac{1}{w(k) + \alpha} \right) - \frac{k w'(k)}{w(k) + \alpha}. \quad (4.8)$$

We now prove the following claim: $kw(0) - \frac{w'(0)}{4}[w(k) - w(0)] \geq 0$ for $k \in (0, k^*)$. Indeed,

$$kw(0) - \frac{w'(0)}{4}[w(k) - w(0)] = \left(1 - \frac{\rho^2 \theta \varphi^2}{8} \right) \theta k + \frac{\rho \varphi \theta^2}{8} - \frac{\rho \varphi \theta^2}{8} \sqrt{\varphi^2 k^2 + 2\varphi \rho k + 1}.$$

Condition 2 of Theorem 4.2 implies that $1 - \frac{\rho^2 \theta \varphi^2}{8} \geq 0$. Then (recall that $\rho \leq 0$) the right-hand side of the above equality represents an increasing function on $(0, k^*)$ which is equal to zero at the origin, and the claim holds. Then, from (4.8),

$$\begin{aligned} g_\alpha(k) - g_\alpha(0) &\geq \frac{-w'(k)}{(w(0) + \alpha)(w(k) + \alpha)} \left\{ k(w(0) + \alpha) - \frac{w'(0)}{4} [w(k) - w(0)] \right\} \\ &\geq 0. \end{aligned}$$

□

Remark 4.6. Given a set of expirations $0 < t_1 < \dots < t_n$ ($n \geq 1$) and at-the-money implied total variances $0 < \theta_{t_1} < \dots < \theta_{t_n}$, Corollary 4.1 gives us the freedom to match three features of one smile (level, skew, and curvature say) but only two features of all the other smiles (level and skew say), subject of course to the given smiles being themselves arbitrage-free. Theorem 4.3 may allow us to match an additional feature of each smile through α_t .

5 Numerics and calibration methodology

5.1 How to eliminate butterfly arbitrage

In Section 4, we showed how to define a volatility smile that is free of butterfly arbitrage. This smile is completely defined given three observables. The ATM volatility and ATM skew are obvious choices for two of them. The most obvious choice for the third observable in equity markets would be the asymptotic slope for k negative and in FX markets and interest rate markets, perhaps the ATM curvature of the smile might be more appropriate.

In view of Lemma 4.1, supposing we choose to fix the SVI-JW parameters v_t , ψ_t and p_t of a given SVI smile, we may guarantee a smile with no butterfly arbitrage by choosing the remaining parameters c'_t and $\tilde{v}'_t$ as

$$c'_t = p_t + 2\psi_t, \quad \text{and} \quad \tilde{v}'_t = v_t \frac{4p_t c'_t}{(p_t + c'_t)^2}.$$

In other words, given a smile defined in terms of its SVI-JW parameters, we are guaranteed to be able to eliminate butterfly arbitrage by changing the call wing c_t and the minimum

variance $\tilde{v}_t$, both parameters that are hard to calibrate with available quotes in equity options markets.

Example 5.1. Consider again the arbitrageable smile from Example 3.1. The corresponding SVI-JW parameters read

$$(v_t, \psi_t, p_t, c_t, \tilde{v}_t) = (0.01742625, -0.1752111, 0.6997381, 1.316798, 0.0116249).$$

We know then that choosing $(c_t, \tilde{v}_t) = (c_t^o, \tilde{v}_t^o) := (0.3493158, 0.01548182)$ gives a smile free of butterfly arbitrage. It follows by continuity of the parameterization in all of its parameters, that there must exist some pair of parameters $(c_t^*, \tilde{v}_t^*)$ with $c_t^* \in (c_t^o, c_t)$ and $\tilde{v}_t^* \in (\tilde{v}_t, v_t^o)$ such that the new smile is free of butterfly arbitrage and is as close as possible to the original one in some sense. In this particular case, choosing the objective function as the sum of squared option price differences plus a large penalty for butterfly arbitrage, we arrive at the following “optimal” choices of the call wing and minimum variance parameters that still ensure no butterfly arbitrage:

$$(c_t^*, \tilde{v}_t^*) = (0.8564763, 0.0116249).$$

Note that the optimizer has left $\tilde{v}_t$ unchanged but has decreased the call wing. The resulting smiles and plots of the function g are shown in Figure 2.

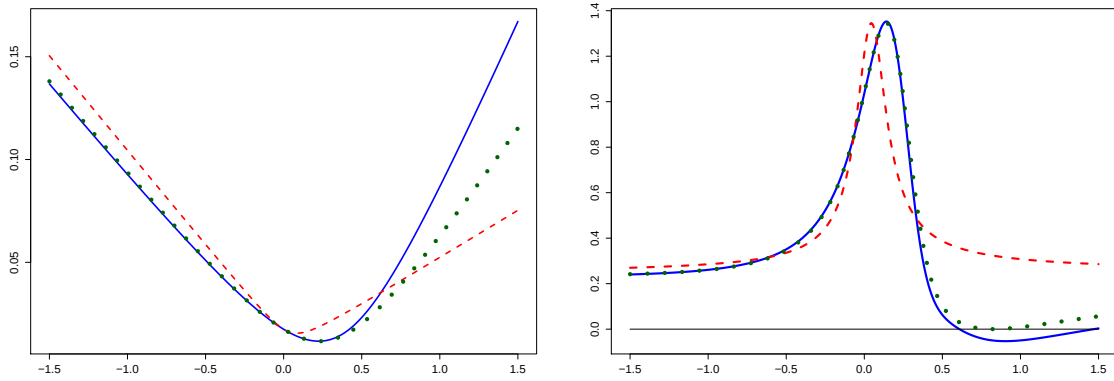


Figure 2: Plots of the total variance smile (left) and the function g defined in (2.1) (right), using the parameters (3.8). The graphs corresponding to the original Vogt parameters is solid, to the guaranteed butterfly-arbitrage-free parameters dashed, and to the “optimal” choice of parameters dotted.

Remark 5.1. The additional flexibility potentially afforded to us through the parameter α_t of Theorem 4.3 sadly does not help us with the Vogt smile of Example 5.1. For α_t to help, we must have $\alpha_t > 0$; it is straightforward to verify that this translates to the condition $v_t(1 - \rho^2) < \tilde{v}_t$ which is violated in the Vogt case.

5.2 Calibration of SVI parameters to implied volatility data

There are many possible ways of defining an objective function, the minimization of which would permit us to calibrate SVI to observed implied volatilities. Whichever calibration strategy we choose, we need an efficient fitting algorithm and a good choice of initial guess. The approach we will present here involves taking a square-root fit as the initial guess. We then fit SVI slice-by-slice with a heavy penalty for calendar spread arbitrage (i.e. crossed lines on a total variance plot). Consider two SVI slices with parameters χ_1 and χ_2 where $t_2 > t_1$. We first compute the points k_i ($i = 1, \dots, n$) with $n \leq 4$ at which the slices cross, sorting them in increasing order. If $n > 0$, we define the points $\tilde{k}_i$ as

$$\begin{aligned}\tilde{k}_1 &:= k_1 - 1, \\ \tilde{k}_i &:= \frac{1}{2}(k_{i-1} + k_i), \quad \text{if } 2 \leq i \leq n, \\ \tilde{k}_{n+1} &:= k_n + 1.\end{aligned}$$

For each of the $n + 1$ points $\tilde{k}_i$, we compute the amounts c_i by which the slices cross:

$$c_i = \max \left[0, w(\tilde{k}_i; \chi_1) - w(\tilde{k}_i; \chi_2) \right].$$

Definition 5.1. *The crossedness of two SVI slices is defined as the maximum of the c_i ($i = 1, \dots, n$). If $n = 0$, the crossedness is null.*

An example SVI calibration recipe

- Given mid implied volatilities $\sigma_{ij} = \sigma_{BS}(k_i, t_j)$, compute mid option prices using the Black-Scholes formula.
- Fit the square-root SVI surface by minimizing sum of squared distances between the fitted prices and the mid option prices. This is now the initial guess.
- Starting with the square-root SVI initial guess, change SVI parameters slice-by-slice so as to minimize the sum of squared distances between the fitted prices and the mid option prices with a big penalty for crossing either the previous slice or the next slice (as quantified by the crossedness from Definition 5.1).

There are obviously many possible variations on this recipe. The objective function may be changed and when finally working to optimize the fit slice-by-slice, one can work from the shortest expiration to the longest expiration or in the reverse order. In practice, working forward or in reverse seems to make little difference. Changing the objective function on the other hand will make some difference especially for very short expirations.

5.3 Interpolation and extrapolation of calibrated slices

Suppose we follow the above recipe above to fit SVI to options with a discrete set of expiries. In particular, each of the resulting SVI smiles will be free of butterfly arbitrage.

It's not immediately obvious that we can interpolate these smiles in such a way as to ensure the absence of static arbitrage in the interpolated surface. The following lemma shows that it is possible to achieve this.

Lemma 5.1. *Given two volatility smiles $w(k, t_1)$ and $w(k, t_2)$ with $t_1 < t_2$ where the two smiles are free of butterfly arbitrage and such that $w(k, \tau_2) \geq w(k, \tau_1)$ for all k , there exists an interpolation such that the interpolated volatility surface is free of static arbitrage for $t_1 < t < t_2$.*

Proof. Given the two smiles $w(k, t_1)$ and $w(k, t_2)$, we may compute the (undiscounted) prices $C(F_i, K_i, t_i) =: C_i$ of European calls with expirations t_i ($i = 1, 2$) using the Black-Scholes formula. In particular, since the two smiles are free of butterfly arbitrage,

$$\frac{\partial^2 C_i}{\partial K^2} \geq 0, \quad \text{for } i = 1, 2.$$

Consider any monotonic interpolation θ_t of the at-the-money implied total variance $w(0, t)$. Let $K_i = F_i e^k$ and $K_t = F_t e^k$. Then for any $t_1 < t < t_2$, define the price $C_t = C(F_t, K_t, t)$ of a European call option to be

$$\frac{C_t}{K_t} = \alpha_t \frac{C_1}{K_1} + (1 - \alpha_t) \frac{C_2}{K_2}, \quad (5.1)$$

where for any $t \in (t_1, t_2)$, we define

$$\alpha_t := \frac{\sqrt{\theta_{t_2}} - \sqrt{\theta_t}}{\sqrt{\theta_{t_2}} - \sqrt{\theta_{t_1}}} \in [0, 1]. \quad (5.2)$$

By construction, for fixed k , the inequality

$$\frac{\partial}{\partial \tau} \frac{C_t}{K_t} \geq 0$$

holds so that there is no calendar spread arbitrage. Also, because of the square-roots in the definition (5.2), the at-the-money interpolated option price will be almost perfectly consistent with the chosen implied total variance interpolation θ_t . Moreover, if the two smiles $w(k, t_1)$ and $w(k, t_2)$ are free of butterfly arbitrage, we have $\partial_{K,K} C(k, t) \geq 0$. To see this, first note that because all the options have the same moneyness, the identity (5.1) is equivalent to

$$\frac{C_t}{F_t} = \alpha_t \frac{C_1}{F_1} + (1 - \alpha_t) \frac{C_2}{F_2}. \quad (5.3)$$

Then note that the ratio $C(F, K, t)/F$ is a function of F and K only through the log-moneyness k . Also, for $K = K_t, K_1, K_2$, we have

$$K^2 \frac{\partial^2 f}{\partial K^2} = \frac{\partial^2 f}{\partial k^2} - \frac{\partial f}{\partial k}.$$

Applying this to (5.3), we obtain

$$\frac{K_\tau^2}{F_t} \frac{\partial^2 C_t}{\partial K_t^2} = \alpha_t \frac{K_1^2}{F_1} \frac{\partial^2 C_1}{\partial K_1^2} + (1 - \alpha_t) \frac{K_2^2}{F_2} \frac{\partial^2 C_2}{\partial K^2}.$$

All the terms on the rhs are non-negative, so the lhs must also be non-negative. We conclude that there is no butterfly arbitrage in the interpolated slice and thus that there is no static arbitrage. The interpolated volatility surface may be retrieved by inversion of the Black-Scholes formula. $\square$

We could conceive of a myriad of algorithms for extrapolating the volatility surface. For example, one way to extrapolate a given set of $n \geq 1$ (arbitrage-free) volatility smiles with expirations $0 < t_1 < \dots < t_n$ would be as follows: At time $t_0 = 0$, the value of a call option is just the intrinsic value. We may then interpolate between t_0 and t_1 using the algorithm presented in the proof of Lemma 5.1, thereby guaranteeing no static arbitrage. For extrapolation beyond the final slice, we suggest to first recalibrate the final slice using the SSVI form (4.1). Then fix a monotonic increasing extrapolation of θ_t (asymptotically linear in time would seem to be reasonable) and extrapolate the smile for $t > t_n$ according to

$$w(k, \theta_t) = w(k, \theta_{t_n}) + \theta_t - \theta_{t_n},$$

which is free of static arbitrage if $w(k, \theta_{t_n})$ is free of butterfly arbitrage by Theorem 4.3.

5.4 A calibration example

We take SPX option quotes as of 3pm on 15-Sep-2011 (the day before triple-witching) and compute implied volatilities for all 14 expirations. The result of fitting square-root SVI is shown in Figure 3. The result of fitting SVI following the recipe provided in Section 5.2 is shown in Figure 4. With the sole exception of the first expiration (options expiring at the market open on the following morning), the fit quality is almost perfect.

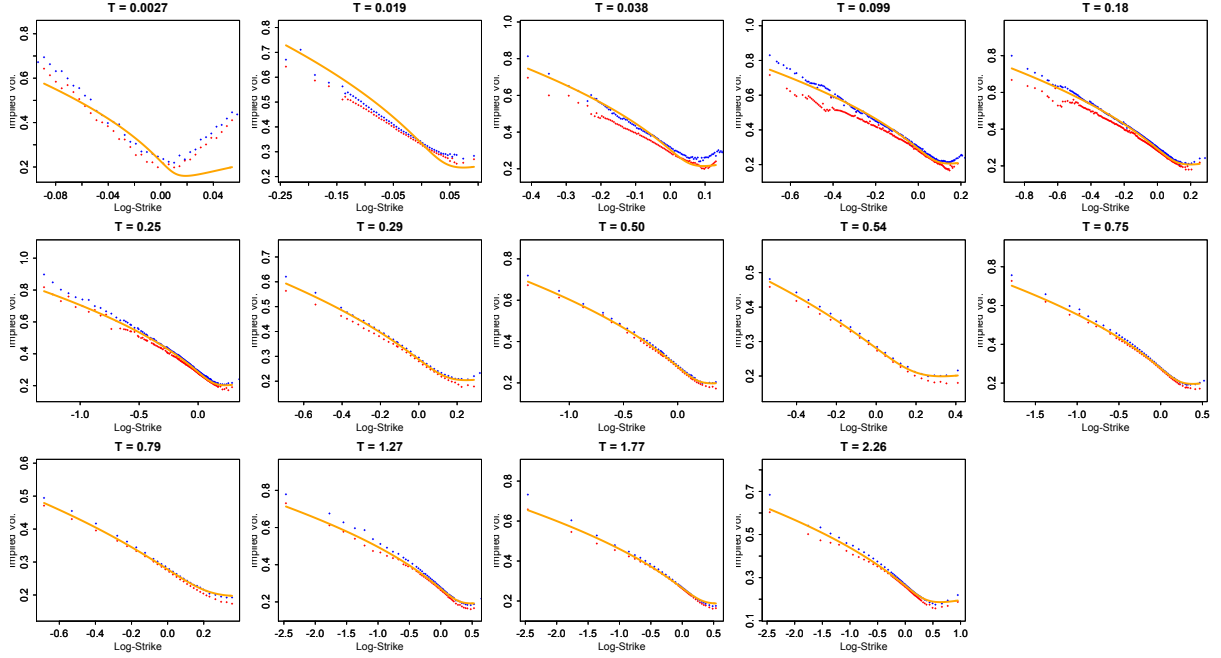


Figure 3: Red dots are bid implied volatilities; blue dots are offered implied volatilities; the orange solid line is the square-root SVI fit

6 Summary and conclusion

We have found and described a large class of arbitrage-free SVI volatility surfaces with a simple closed-form representation. Taking advantage of the existence of such surfaces, we showed how to eliminate both calendar spread and butterfly arbitrages when calibrating SVI to implied volatility data. We have also demonstrated the high quality of typical SVI fits with a numerical example using recent SPX options data. The potential applications of this work to modelling the dynamics of the implied volatility surface are left for future research.

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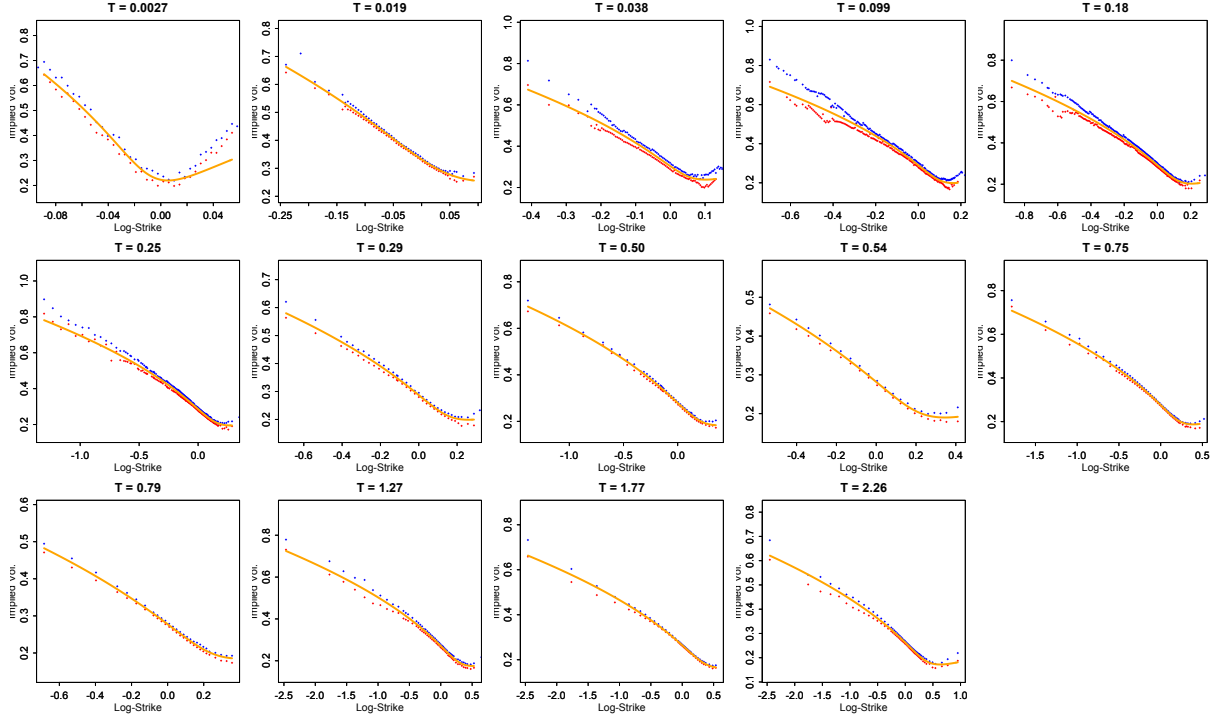


Figure 4: Red dots are bid implied volatilities; blue dots are offered implied volatilities; the orange solid line is the SVI fit following recipe of Section 5.2

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